Worksheet A: Cups, Dice and Data

Name: _____ Date: _____

Task 1: Monty Hall in pairs

in class, pairs, 10 min.

The host secretly rolls a die to place the ball (1–2: A, 3–4: B, 5–6: C), the player points at a cup, and the host lifts an *empty* cup that was not chosen. Odd rounds: stay; even rounds: switch. Tick whether the round was won.

Rd.	Strategy	Ball	won	Rd.	Strategy	Ball	won
1	stay			13	stay		
2	switch			14	switch		
3	stay			15	stay		
4	switch			16	switch		
5	stay			17	stay		
6	switch			18	switch		
7	stay			19	stay		
8	switch			20	switch		
9	stay			21	stay		
10	switch			22	switch		
11	stay			23	stay		
12	switch			24	switch		

Total: **stay** won _____ of _____ **switch** won _____ of _____ whole class (board): _____

Task 2: Dice detectives

in class, pairs, 10 min.

The roller opens the envelope and reads the card *privately*: "fair" (only a 6 counts as a hit) or "loaded" (5 and 6 count as hits). The roller throws the die 30 times, hidden under a cup, and only says "hit" or "no hit". The detective keeps the tally. Known in advance: one envelope in five is loaded.

Hit (tally)	
No hit (tally)	

Hits k	3	4	5	6	7	8	9	10	11	12	13	14
$p = P(X \geq k \mid \text{fair})$	0.90	0.76	0.58	0.38	0.22	0.11	0.051	0.020	0.007	0.002	0.0005	0.0001
$BF_{10} = 2^k \cdot 0.8^{30-k}$	0.02	0.05	0.12	0.30	0.76	1.9	4.7	11.8	30	74	184	461
$P(\text{loaded})$ at $1:4$	0.5 %	1 %	3 %	7 %	16 %	32 %	54 %	75 %	88 %	95 %	98 %	99 %

1. **Frequentist:** H_0: fair. p-value for your k: _____. At $\alpha = 0.05$ the die is ☐ significantly loaded ☐ not significant.
2. **Bayesian:** BF_{10} = _____; prior odds $1:4$, posterior odds _____; $P(\text{loaded})$ = _____. Do the calculation once yourself, without the last row of the table.
3. Open the envelope. Who was right? Enter your pair in the two-by-two table on the board.
4. What would your Bayesian verdict be if only one die in 100 were loaded? And your test verdict?

Task 3: Analysing a data set

homework, small groups

Fictitious data set (constructed for this exercise, not a real survey): 50 people held an ideomotor pendulum (cf. Unit 1). The largest swing was measured in millimetres (**mm**); afterwards the people rated how amazed they were (**W** for wonder, 1 = not at all to 5 = very much) and stated whether they had known the pendulum effect before (**PK** for prior knowledge, 1 = yes, 0 = no).

No	mm	W	PK	No	mm	W	PK	No	mm	W	PK	No	mm	W	PK	No	mm	W	PK
1	11	3	0	11	9	3	0	21	10	2	1	31	43	3	1	41	47	4	0
2	5	2	0	12	24	2	0	22	26	4	1	32	57	4	0	42	59	3	0
3	15	3	1	13	24	4	1	23	39	4	0	33	21	3	0	43	26	3	0
4	41	4	1	14	8	2	0	24	15	4	0	34	96	5	0	44	31	4	1
5	37	3	0	15	16	2	1	25	13	3	0	35	49	4	0	45	23	2	0
6	39	5	0	16	25	4	0	26	13	3	0	36	30	4	0	46	20	3	1
7	18	3	0	17	19	5	0	27	66	5	0	37	23	4	0	47	12	1	1
8	0	1	0	18	53	4	1	28	35	3	1	38	53	4	0	48	36	4	1
9	16	2	1	19	5	2	1	29	38	5	0	39	18	2	0	49	18	3	0
10	31	4	1	20	45	5	0	30	26	3	1	40	73	5	0	50	40	4	0

1. State the level of measurement of each variable.
2. For **mm**, compute the minimum, maximum, range, median, Q_1, Q_3, *IQR* and mean. State how you determined the quartiles.
3. Draw a histogram (bin width 10 mm) and a box plot. Are there outliers according to the $1.5 \cdot IQR$ rule (values more than $1.5 \cdot IQR$ below Q_1 or above Q_3)? What would you do with them?
4. Compute the mean and median of **mm** without the outlier. Which statistic reacts more strongly, and why?
5. Summarise **W** in a sensible way. Is a mean admissible?
6. Compare **mm** between people with and without prior knowledge, using the mean *and* the median. What do you notice?
7. For the same comparison, compute a t-test with the pooled standard deviation and the Bayes factor from the rule of thumb (Wagenmakers 2007), where N is the total number of people:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{1/n_1 + 1/n_2}}, \quad df = n_1 + n_2 - 2, \qquad BF_{01} \approx \sqrt{N} \left(1 + \frac{t^2}{df}\right)^{-N/2}$$

   What do the two results say, and what do they not say?